Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Total derivatives, Chain rule

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Total derivatives, Chain rule

1 Problems on Total derivatives, Chain rule

Study focus: Problems on Total derivatives, Chain rule. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000054 · EV0000055

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on total derivative, chain rule, and implicit function.

Key details

- Point du by dt at a point t equals to pi by 2 where u equals x square y times a power x y and x equals t square plus 1 and y equals sine t.
- From the data, we can see that u is a function of x and y, where x and y are the functions of p.
- Now because we want du by dt at the point t equals to pi by 2, we need to find.
- X and Y at the point.
- At t equals to pi by 2, x becomes pi square by 4 plus 1 and y becomes 1 because of sine t.
- Also, du by dt becomes because y is 1, it becomes only e power x times 2x plus x square.
- We also see that when t is pi by 2, the second term doesn't exist because d by dt is cost t and the cost t is 0 at t equals to pi by 2.
- Pi square by four plus one of two times pi square by four plus one plus pi square by four plus one multiply.
- This is the value of d u by dt at the point d equals to pi by two.
- Find du by dt.
- We use a same approach what we have used for the previous problem.
- Plus two y divided by y square plus one.
- times second square t minus x e power x y plus two y divided by y square plus one into sin t.
- Now because u is a function of t, we express x and y in terms of t.
- The base of a triangle is increasing at a rate of 2 centimeter per second, and its height is increasing at a rate of 1 centimeter per second.
- At a certain instant, the base is 10 centimeters, the height is 6 centimeters.

Example or application from the lecture

- Let's begin with total derivatives.
- For the given x and y, d x by dt as well as dy by dt becomes 2t as well as cos t respectively.
- Let us look at a similar problem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 05:00–09:59

Find the rate at which the area of a triangle is increasing at that instant.

Key details

- And the height of a triangle is increasing at a rate of 1 centimeter per second.
- Also, p and h are respectively 10 and 6.
- It means the rate at which the area of the triangle is increasing is 11 square centimeters per second.
- Water is being poured into a conical tank at a rate of 9 cubic centimeter per second.
- Find the rate at which the water level is rising when the height is 6 centimeter.
- pi r square h.
- Because height and radius are equal when we replace r equals h, v becomes 1 by 3 pi h power 3.
- After calcation 3 and 3, we get dv by dt to be pi h square dh by dt.
- P V by D T is 9 and H is 6.
- It means u is a function of x and y, where x and y are the functions of s and t.
- then u becomes composite function of s and t.
- Do x by dou s is going to be t times e power s t. dou x by dou t becomes s times e power s t. becomes two s dou y by dou t becomes two t.
- Then find a value of douz by doux as well as douz by dou y at the point zero.
- We know from theory the formula for dou z by doux is minus of f x divided by f z.
- And similarly, dou z by dou y equals minus f y divided by f z.
- It means we take this as some function of xy and z, then we find only partial derivative of f with respect to x y.

Example or application from the lecture

- In the problem, it is given that the base of a triangle is increasing at a rate of 2 centimeters per second.
- Let us look at one more application problem on total derivative.
- It is given that the height and radius of the cone are equal.

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EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 10:00–14:59

as well as z.

Key details

- Then using the formula we can compute douz by doux and dou z by dou y at the point zero.
- Of course there are two ways to solve this problem.
- Another method what we can use we can use dy by dx equals.
- Minus of partial derivative of f with respect to x divided by f with respect to y.
- It means we can take this as some function of x and y, then use the partial derivative and use this formula, we can get the answer.
- So another method which is called direct method, which is shown as follows.
- Data x square plus xy plus y square equal to 7.
- Differentiate both the sets with respect to x.

Example or application from the lecture

- Let us solve another problem on implicit function with two variables.
- If the relation between x and y is given by x square plus x pi plus y square equals 7.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Total derivatives, Chain rule transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Total derivatives, Chain rule transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00014	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...
SRC00015	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...
SRC00017	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare...

C Revision and retrieval practice